

Business Case

Organization and department

St. Clair College, Zekelman School of Business

Requirements and Justification

Impact of Online and Offline Education on Student Performance and Engagement

Executive Summary

This project aims to assess the impact of online and offline education on student performance, engagement, satisfaction, and retention. With the rise of online learning platforms accelerated by the COVID-19 pandemic, educational institutions are increasingly faced with questions about the effectiveness of online learning compared to traditional in-person education. The study will collect and analyze data to understand which educational mode is more beneficial in specific contexts, helping to inform strategic decisions about future learning models at St. Clair College.

Functional Requirements

- Conduct an online survey through Google Forms distributed to St. Clair students via email and WhatsApp to collect data on student experiences.
- Perform data cleaning, integration, and exploratory data analysis (EDA) to ensure accuracy and reliability of insights.
- Utilize machine learning models for predictive analysis to understand potential trends in student behavior and performance in online and offline contexts.
- Create data visualizations using Tableau to communicate insights effectively.
- Produce a comprehensive report detailing findings, conclusions, and actionable

Non-Functional Requirements

- **Reliability:** Ensuring high data quality and analysis accuracy to support decision-making.
- **Scalability:** The project framework and methods should be scalable to accommodate future studies or data sources.
- **Usability:** The final report and visualizations should be easily interpretable for stakeholders without technical backgrounds.

Solution and cost estimation

Resources	Cost
Google Forms, Canva, Photoshop	\$500
Tableau Software (license)	\$1200

Data analysis tools and software	\$1500
Personnel costs (data analysts, researchers)	\$10,000
Total	\$13,200

Benefits of investing in this solution

1. **Informed Strategic Planning:** By identifying which educational mode is more effective in specific contexts, St. Clair College can develop tailored strategies that boost student performance and engagement.
2. **Improved Resource Allocation:** Insights into student preferences and challenges will allow the College to allocate resources effectively, enhancing both online and offline learning environments.
3. **Enhanced Student Satisfaction and Retention:** Understanding student needs and addressing challenges in online and offline education can improve overall satisfaction, retention, and academic success rates.

Expected Outcomes

1. Identification of differences in student performance and engagement between online and offline education.
2. Insights into key factors influencing student preferences for online or offline learning modes.
3. Analysis of demographic influences on student outcomes in each mode.
4. Recommendations for optimizing both online and offline education, based on identified challenges and strengths.